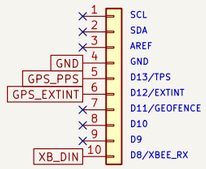
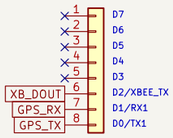
J1



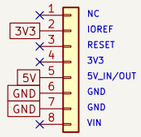
ArduSimple SCL..D8

J2



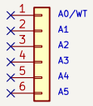
ArduSimple D7..D0

J3



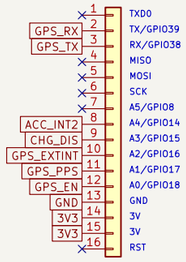
ArduSimple power

J4



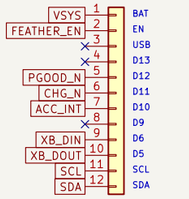
ArduSimple A0..A5

J5



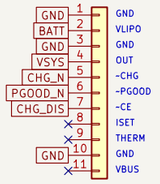
Feather left

J6



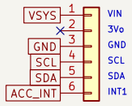
Feather right

J7



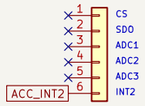
bq24074 (soldered)

J8



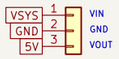
LIS3DH (soldered)

J9



LIS3DH (soldered, support)

J10



S9V11F5 (soldered)

J11

Battery JST-PH

JP1

Charger bypass: shunt only without bq24074

TP1



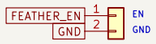
GND

TP2



GPS_EN wire to S9V11F5 EN

J12



Feather off: short EN to GND

The ArduSimple and the Feather plug into sockets; the LIS3DH, S9V11F5 and bq24074 are soldered low on their own headers. Nets join by label name.

ArduSimple: male headers soldered underneath; power switch DOWN (5V IN, IOREF N/C); --role rover (UART2 off).

Feather on 8.5 mm stacking headers (housings on the display side), tails into J5/J6; display up, USB-C at the front. LIS3DH runs from VSYS (its own 3.3 V regulator).

bq24074 optional: fitted -> JP1 open, battery feeds the charger, charger OUT feeds VSYS.

No bq24074 -> fit the JP1 shunt (BATT to VSYS). Never both: the shunt would bypass the charger.

~CHG and ~PGOOD are open drain to Feather D11 / D12 (read with INPUT_PULLUP); the firmware does not use them yet.

Optional, unused by the firmware so far: TIMEPULSE -> A1, EXTINT <- A2, bq ~CE <- A3 (high stops charging; 10k pull-down on the bq), LIS3DH INT2 -> A4.

GPS_EN: Feather A0 to pad TP2; wire TP2 to the S9V11F5 EN hole. Build with -DGPS_POWER_PIN=18 to cut the GPS and XBee in deep sleep.

J12: short EN to GND to switch the Feather off (the regulator and ArduSimple stay on battery power).

Feather S3 Reverse TFT, ArduSimple Budget, LIS3DH, S9V11F5, optional bq24074

race-tracker

Sheet: /

File: esp32-rover-carrier.kicad_sch

Title: ESP32 rover carrier

Size: A3	Date:	**Rev: C**
KiCad E.D.A. 10.0.8		Id: 1/1